

Theorem (Order properties of $\mathbb{R}$): We have

- Trichotomy: for all $x, y \in \mathbb{R}$ exactly one of $x < y$ or $x = y$ or $x > y$ is true.
- Transitivity: for all $x, y, z \in \mathbb{R}$, if $x < y$ and $y < z$, then $x < z$.
- Compatibility with $+$: for all $x, y, z \in \mathbb{R}$, if $x < y$, then $x + z < y + z$.
- Compatibility with $\times$: for all $x, y, z \in \mathbb{R}$, if $x < y$ and $0 < z$, then $z \times x < z \times y$.

Definition (Supremum): Let S be a subset of $\mathbb{R}$ that is bounded above. A number x is said to be a supremum or least upper bound of S if

- x is an upper bound of S .
- for any y , if y is an upper bound of S , then $x \leq y$.

Axiom (Completeness axiom): For any non-empty subset S of $\mathbb{R}$, if S has an upper bound, then S has a supremum.

Theorem (Archimedean Property of $\mathbb{R}$): For any $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $x < n$.

Definition (Infimum): Let S be a subset of $\mathbb{R}$ that is bounded below. A number x is said to be a infimum or greatest lower bound of S if

- x is a lower bound of S .
- for any y , if y is a lower bound of S , then $y \leq x$.